

Vector Calculus

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§1 Preliminaries

§1.1 Vector notation

We will in general work in $\mathbb{R}^3$, but sometimes $\mathbb{R}^2$. In fact, most of our results will generalise to $\mathbb{R}^n$.

We will write column vectors to save space when we mean row vectors, and we will use dot notation for differentiation (even though t isn't really time).

If $\mathbf{a}$ and $\mathbf{b}$ are vectors, then the following hold:

$$\dot{\mathbf{a}}(t) = \frac{d\mathbf{a}}{dt} = \frac{d}{dt}(a_1, a_2, a_3) = (\dot{a}_1, \dot{a}_2, \dot{a}_3) \quad (1)$$

$$\frac{d}{dt}(\mathbf{a} \cdot \mathbf{b}) = \dot{\mathbf{a}} \cdot \mathbf{b} + \mathbf{a} \cdot \dot{\mathbf{b}} \quad (2)$$

$$\frac{d}{dt}(\mathbf{a} \times \mathbf{b}) = \dot{\mathbf{a}} \times \mathbf{b} + \mathbf{a} \times \dot{\mathbf{b}} \quad (3)$$

and these results can be shown using suffix notation. Integration is also componentwise. We have therefore

$$\int_{t_1}^{t_2} \dot{\mathbf{a}}(t) dt = \mathbf{a}(t_2) - \mathbf{a}(t_1)$$

The modulus of a vector will be denoted by the same letter as the vector, but not bolded/underlined. So, for example, $r = |\mathbf{r}|$ (but x means x_1 , not $|\mathbf{x}|$).

Notationally, we might write ∂_i for $\frac{\partial}{\partial x_i}$.

§1.2 The chain rule

Recall that, if f is a function of $\mathbf{x}$ which is itself a function of t , then

Proposition 1.1

$$\begin{aligned} \frac{d}{dt}f(\mathbf{x}(t)) &= \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt} \\ &= \frac{\partial f}{\partial x_i} \frac{dx_i}{dt} \end{aligned}$$

If instead $\mathbf{x} = \mathbf{x}(\mathbf{u})$, where $\mathbf{u} = (u_1, u_2, u_3) = (u, v, w)$, then

$$\frac{\partial f}{\partial u} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial u} + \frac{\partial f}{\partial z} \frac{\partial z}{\partial u}$$

and similarly for $\frac{\partial f}{\partial v}$ and $\frac{\partial f}{\partial w}$, i.e.

$$\frac{\partial f}{\partial u_i} = \frac{\partial f}{\partial x_i} \frac{\partial x_i}{\partial u_i}$$

§1.3 Taylor's theorem

Recall that for a small vector displacement $\delta \mathbf{x}$,

$$\begin{aligned} f(\mathbf{x} + \delta \mathbf{x}) &= f(x + \delta x, y + \delta y, z + \delta z) \\ &= f(x, y, z) + \frac{\partial f}{\partial x} \delta x + \frac{\partial f}{\partial y} \delta y + \frac{\partial f}{\partial z} \delta z + \frac{1}{2!} \left(\delta x \frac{\partial}{\partial x} + \delta y \frac{\partial}{\partial y} + \delta z \frac{\partial}{\partial z} \right)^2 f + \dots \end{aligned}$$

or, in suffix notation,

Proposition 1.2

$$f(\mathbf{x} + \delta \mathbf{x}) = f(\mathbf{x}) + \frac{\partial f}{\partial x_i} \delta x_i + O(|\delta \mathbf{x}|^2)$$

§1.4 Infinitesimals

In one dimension, we have from Taylor's theorem

$$\delta f = f(x + \delta x) - f(x) = f'(x) \delta x + O(\delta x^2)$$

Taking $\delta x \rightarrow 0$, δx and δf become dx and df , and we get the exact equality

$$df = f'(x) dx$$

If we want, we can even think of infinitesimals as numbers such that their squares are zero.

Note that infinitesimals have their own product and quotient rules, namely

$$\begin{aligned} d(fg) &= g df + f dg \\ d\left(\frac{f}{g}\right) &= \frac{g df - f dg}{g^2} \end{aligned}$$

We also get

Proposition 1.3

$$df = \frac{\partial f}{\partial x_i} dx_i$$

The modulus of an infinitesimal vector can be calculated using, for example

$$|d\mathbf{x}| = |dx| \sqrt{1 + \left(\frac{dy}{dx}\right)^2 + \left(\frac{dz}{dx}\right)^2}$$

If dx and dy are independent infinitesimals then $dx dy$ is an infinitesimal area. But if y depends on x in some way then $dx dy$ vanishes, since it's proportional to dx^2 .

§2 Vector differential operators

Remark. What is an operator? It is a map between vector spaces of functions.

§2.1 Del

The fundamental differential operator is *del*:

$$\nabla = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix}$$

We can also write

$$\nabla = \mathbf{e}_1 \frac{\partial}{\partial x_1} + \mathbf{e}_2 \frac{\partial}{\partial x_2} + \mathbf{e}_3 \frac{\partial}{\partial x_3} = \mathbf{e}_i \frac{\partial}{\partial x_i}$$

where $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ is the canonical basis for $\mathbb{R}^3$.

§2.2 Gradient

When del is applied to a scalar field $f(\mathbf{x})$, it is known as the *gradient* operator:

$$\nabla f = \begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \\ \frac{\partial f}{\partial z} \end{pmatrix}$$

This can also be denoted $\text{grad} f$.

In suffix notation, we have

$$\begin{aligned} (\nabla f)_i &= \frac{\partial f}{\partial x_i} \\ \iff \nabla f &= \frac{\partial f}{\partial x_i} \mathbf{e}_i \end{aligned}$$

Example • If $f(\mathbf{x}) = |\mathbf{x}|^2 = x^2 + y^2 + z^2$, then $\nabla f = (2x, 2y, 2z) = 2\mathbf{x}$. Alternatively, $\frac{\partial}{\partial x_i}(x_j x_j) = 2x_i \delta_{ij} = 2x_i$ (the δ_{ij} comes from $\frac{\partial x_j}{\partial x_i}$).

- $\nabla |\mathbf{x} - \mathbf{x}_0|^2 = 2(\mathbf{x} - \mathbf{x}_0)$. This is a direct consequence of the above and a very simple application of the chain rule.
- $\nabla(\mathbf{a} \cdot \mathbf{x}) = \nabla(a_1 x + a_2 y + a_3 z) = a_1 \mathbf{e}_1 + a_2 \mathbf{e}_2 + a_3 \mathbf{e}_3 = \mathbf{a}$. Alternatively, $\frac{\partial}{\partial x_i}(a_j x_j) = a_i \delta_{ij} = a_i$.

∇f in 3D is analogous to f' in 1D, and we can evaluate it at points.

Example

If $f(\mathbf{x}) = x + y^2 + z^4$, then $\nabla f(\mathbf{a}) = \nabla f(\mathbf{b})$ if and only if $\mathbf{a} - \mathbf{b}$ is parallel to the x -axis. Observe $\nabla f(\mathbf{x}) = (1, 2y, 4z^3)$, so

$$\begin{aligned} (1, 2a_2, 4a_3^3) &= (1, 2b_2, 4b_3^3) \\ \iff (a_2, a_3) &= (b_2, b_3) \\ \iff \mathbf{a} - \mathbf{b} &= (a_1 - b_1, 0, 0) \end{aligned}$$

We have some new opportunities for notation now – for example, we can rewrite (1.1) as

Proposition 2.1

$$\frac{d}{dt}f(\mathbf{x}(t)) = \dot{\mathbf{x}} \cdot \nabla f$$

and (1.3) as

Proposition 2.2

$$df = \nabla f \cdot d\mathbf{x}$$

Equation (2.2) also gives a way of defining ∇f that is *coordinate-independent*, i.e. it doesn't require us to choose a particular basis at any point.

Example

If $f(x) = |\mathbf{x}|^2$, then

$$\begin{aligned} df &= f(\mathbf{x} + d\mathbf{x}) - f(\mathbf{x}) = \mathbf{x} \cdot \mathbf{x} + 2\mathbf{x} \cdot d\mathbf{x} + d\mathbf{x} \cdot d\mathbf{x} - |\mathbf{x}|^2 \\ &= 2\mathbf{x} \cdot d\mathbf{x} \end{aligned}$$

so it must be the case that $\nabla f = 2\mathbf{x}$.

Finally, we may write $\nabla_{\mathbf{x}}$, $\nabla_{\mathbf{y}}$, etc., to denote differentiation w.r.t. a particular vector variable.

§2.3 Divergence

Definition. The *divergence* of a vector field is given by del dotted with the field:

$$\nabla \cdot \mathbf{F} = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \cdot \begin{pmatrix} F_1 \\ F_2 \\ F_3 \end{pmatrix} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} = \frac{\partial F_i}{\partial x_i}$$

More formally, we can derive this expression by

$$\begin{aligned} \nabla \cdot \mathbf{F} &= \left(\mathbf{e}_i \frac{\partial}{\partial x_i} \right) \cdot (F_j \mathbf{e}_j) \\ &= \mathbf{e}_i \cdot \frac{\partial}{\partial x_i} (F_j \mathbf{e}_j) \\ &= \mathbf{e}_i \cdot \frac{\partial F_j}{\partial x_i} \mathbf{e}_j \\ &= \frac{\partial F_j}{\partial x_i} \delta_{ij} = \frac{\partial F_i}{\partial x_i} \end{aligned}$$

What is the divergence? It measures the amount by which a field is ‘expanding’ around $\mathbf{x}$. We will get a better idea later on when we derive a coordinate-independent definition.

Example • Take $F(\mathbf{x}) = \mathbf{x}$. Then $\nabla \cdot F = \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z} = 3$. Suffix notation also works! DIAGRAM

- Take $F(\mathbf{x}) = (-y, x, 0)$. Then $\nabla \cdot F = -\frac{\partial y}{\partial x} + \frac{\partial x}{\partial y} = 0$. DIAGRAM
- If A is a 3 by 3 matrix, we might want to calculate

$$\nabla \cdot (A\mathbf{x}) = \frac{\partial}{\partial x_i} (A_{ij}x_j) = A_{ij}\delta_{ij} = A_{ii} = \text{Tr } A$$

§2.4 Curl

Definition. The *curl* of a vector field is given by del crossed with the field:

$$\nabla \times \mathbf{F} = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \times \begin{pmatrix} F_1 \\ F_2 \\ F_3 \end{pmatrix} = \begin{pmatrix} \frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} \\ \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x} \\ \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \end{pmatrix}$$

In suffix notation, we have

$$(\nabla \times F)_i = \varepsilon_{ijk} \frac{\partial}{\partial x_j} F_k$$

The curl, notably, is only defined in 3D, since we only have a cross product in 3D, unlike gradient or divergence. It measures, in a certain sense, the *rotation* of the field about a point $\mathbf{x}$.

Example • With $F(\mathbf{x}) = \mathbf{x}$, we have

$$\nabla \times F = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \times \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \mathbf{0}$$

Alternatively, $(\nabla \times F)_i = \varepsilon_{ijk} \frac{\partial}{\partial x_j} x_k = \varepsilon_{ijk} \delta_{jk} = 0$.

- With $F(\mathbf{x}) = (-y, x, 0)$, $\nabla \times F = (0, 0, 2)$.
- If A is a 3 by 3 symmetric matrix,

$$(\nabla \times (A\mathbf{x}))_i = \varepsilon_{ijk} \frac{\partial}{\partial x_j} A_{kl}x_l = \varepsilon_{ijk} A_{kj}$$

But ε_{ijk} is antisymmetric while A_{kj} is symmetric, so the terms cancel in pairs and we get $\nabla \times (A\mathbf{x}) = \mathbf{0}$.

§2.5 Directional derivatives

Given a *vector* field $\mathbf{u}$ we can define a scalar *operator*

$$\mathbf{u} \cdot \nabla = u_1 \frac{\partial}{\partial x_1} + u_2 \frac{\partial}{\partial x_2} + u_3 \frac{\partial}{\partial x_3} = u_i \frac{\partial}{\partial x_i}$$

We can then apply this operator to a *scalar* field f :

$$(\mathbf{u} \cdot \nabla)f = u_1 \frac{\partial f}{\partial x_1} + u_2 \frac{\partial f}{\partial x_2} + u_3 \frac{\partial f}{\partial x_3} = u_i \frac{\partial f}{\partial x_i}$$

But, happily, it turns out

$$(\mathbf{u} \cdot \nabla)f = \left(u_i \frac{\partial}{\partial x_i} \right) f = u_i \frac{\partial f}{\partial x_i} = \mathbf{u} \cdot (\nabla f)$$

and so we can omit the brackets.

However, we can also apply $\mathbf{u} \cdot \nabla$ to a *vector* field $\mathbf{F}$:

$$(\mathbf{u} \cdot \nabla)\mathbf{F} = u_1 \frac{\partial}{\partial x_1} \mathbf{F} + u_2 \frac{\partial}{\partial x_2} \mathbf{F} + u_3 \frac{\partial}{\partial x_3} \mathbf{F}$$

so that

$$((\mathbf{u} \cdot \nabla)\mathbf{F})_i = u_j \frac{\partial F_i}{\partial x_j}$$

and this gives us a *vector* field back. Disturbingly, the brackets here *are* important, since $\mathbf{u} \cdot (\nabla \mathbf{F})$ is the gradient of a vector which isn't defined. However, we omit them anyway (since there's no ambiguity).

Example

Take $\mathbf{u} = (0, x, y)$ and $\mathbf{F} = (yz, 0, x)$. Then $\mathbf{u} \cdot \nabla = x \frac{\partial}{\partial y} + y \frac{\partial}{\partial z}$, so $(\mathbf{u} \cdot \nabla)\mathbf{F} = (xz + y^2, 0, 0)$.

Definition. Given a unit vector $\hat{\mathbf{n}}$, the *directional derivative* $D_{\hat{\mathbf{n}}}f$ of f at $\mathbf{x}$ is defined to be the one-dimensional derivative of f along the line from $\mathbf{x}$ parallel to $\hat{\mathbf{n}}$. That is, if we make $g(t) = f(\mathbf{x} + t\hat{\mathbf{n}})$, then

$$D_{\hat{\mathbf{n}}}f(\mathbf{x}) = g'(0) = \hat{\mathbf{n}} \cdot \nabla f(\mathbf{x})$$

by [proposition 2.1](#). Then $D_{\hat{\mathbf{n}}} = \hat{\mathbf{n}} \cdot \nabla$, and so the operator $\mathbf{u} \cdot \nabla$ is also called the directional derivative (even for non-unit vectors $\mathbf{u}$).

§2.6 Level surfaces and the gradient

Definition. In 3D, a *level surface* of a function $f(\mathbf{x})$ is a two-dimensional surface satisfying $f(\mathbf{x}) = c$ for some constant c . In 2D, these are just curves which we refer to as *contours*.

Proposition

The gradient of f is always normal to a level surface.

Proof. Starting at $\mathbf{x}$ on the surface, consider a displacement $d\mathbf{x}$ within the surface. Because $f = c$, $df = 0$ for this displacement, so, by [proposition 2.2](#), $\nabla f \cdot d\mathbf{x} = df = 0$ for all $d\mathbf{x}$ lying in the surface. So ∇f is perpendicular to all of these and hence is perpendicular to the surface. ■

Example

Take $f(\mathbf{x}) = |\mathbf{x}|^2$, so $\nabla f = 2\mathbf{x}$. Since this is simply radially outwards, we conclude the level surfaces are spheres centred at the origin.

Proposition

∇f always points in the direction in which f grows the most rapidly.

Proof. Consider $D_{\hat{\mathbf{n}}}f = \hat{\mathbf{n}} \cdot \nabla f$ for an arbitrary unit vector $\hat{\mathbf{n}}$. Clearly $\hat{\mathbf{n}} \cdot \nabla f$ is maximised when $\hat{\mathbf{n}}$ and ∇f are parallel, and in particular this maximum is $|\nabla f|$. ■

Remark. We can think of level surfaces in two different ways, and endow them with different grads. Suppose we have a ground level $z = h(x, y)$ (imagine hills). Working in 2D, we have that ∇h points in the direction that is steepest uphill (or the projection of that onto the x - y plane). On the other hand, if we consider $f(x, y, z) = z - h(x, y)$ (which defines the exact same surface), then the level surface $f = 0$ has ∇f pointing normal to the surface.

§2.7 Various vector differential identities

The following important identities can be proven using suffix notation.

Proposition 2.3

$$\nabla(fg) = f\nabla g + g\nabla f$$

Proposition 2.4

$$\nabla(f(g(\mathbf{x}))) = f'(g(\mathbf{x}))\nabla g$$

Proposition 2.5

$$\nabla \cdot (f\mathbf{F}) = f\nabla \cdot \mathbf{F} + \mathbf{F} \cdot \nabla f$$

Proof. We have

$$\begin{aligned} \nabla \cdot (f\mathbf{F}) &= \frac{\partial}{\partial x_i}(fF_i) \\ &= \frac{\partial f}{\partial x_i}F_i + f\frac{\partial F_i}{\partial x_i} \\ &= \nabla f \cdot \mathbf{F} + f\nabla \cdot \mathbf{F} \end{aligned}$$

■

Proposition 2.6

$$\nabla \times (f\mathbf{F}) = f\nabla \times \mathbf{F} + \nabla f \times \mathbf{F}$$

The above propositions should be memorised – it is not necessary to memorise the following propositions, but one should be able to produce proofs of them.

Proposition 2.7

$$\nabla(\mathbf{F} \cdot \mathbf{G}) = \mathbf{F} \times (\nabla \times \mathbf{G}) + (\mathbf{F} \cdot \nabla)\mathbf{G} + \mathbf{G} \times (\nabla \times \mathbf{F}) + (\mathbf{G} \cdot \nabla)\mathbf{F}$$

Proof. Counterintuitively, let us start with

$$\begin{aligned} (\mathbf{F} \times (\nabla \times \mathbf{G}))_i &= \varepsilon_{ijk} F_j \varepsilon_{klm} \frac{\partial}{\partial x_l} G_m \\ &= (\delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}) F_j \frac{\partial G_m}{\partial x_l} \\ &= F_j \frac{\partial G_j}{\partial x_i} - F_j \frac{\partial G_i}{\partial x_j} \end{aligned}$$

Adding to a similar expression for $(\mathbf{G} \times (\nabla \times \mathbf{F}))_i$, we get

$$\begin{aligned} &F_j \frac{\partial G_j}{\partial x_i} - F_j \frac{\partial G_i}{\partial x_j} + G_j \frac{\partial F_j}{\partial x_i} - G_j \frac{\partial F_i}{\partial x_j} \\ &= \underbrace{F_j \frac{\partial G_j}{\partial x_i} + G_j \frac{\partial F_j}{\partial x_i}}_{\frac{\partial}{\partial x_i}(F_j G_j)} - F_j \frac{\partial G_i}{\partial x_j} - G_j \frac{\partial F_i}{\partial x_j} \\ &= (\nabla(\mathbf{F} \cdot \mathbf{G}))_i - ((\mathbf{F} \cdot \nabla)\mathbf{G})_i - ((\mathbf{G} \cdot \nabla)\mathbf{F})_i \end{aligned}$$

and the result follows. ■

Proposition 2.8

$$\nabla \cdot (\mathbf{F} \times \mathbf{G}) = -\mathbf{F} \cdot (\nabla \times \mathbf{G}) + \mathbf{G} \cdot (\nabla \times \mathbf{F})$$

Proposition 2.9

$$\nabla \times (\mathbf{F} \times \mathbf{G}) = (\nabla \cdot \mathbf{G})\mathbf{F} - (\nabla \cdot \mathbf{F})\mathbf{G} + (\mathbf{G} \cdot \nabla)\mathbf{F} - (\mathbf{F} \cdot \nabla)\mathbf{G}$$

Remark. We must not use the formulae $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}$ or $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}$, since our derivations assumed certain commutativity and anti-commutativity properties that don't hold for operators.

We have one more very useful identity.

Proposition 2.10

$$\frac{\partial r}{\partial x_i} = \frac{x_i}{r}$$

Proof. We have $r^2 = x_j x_j$, so $2r \frac{\partial r}{\partial x_i} = 2x_j \frac{\partial x_j}{\partial x_i} = 2x_j \delta_{ij} = 2x_i$. ■

Corollary 2.11

$$\nabla f(r) = f'(r) \mathbf{e}_r$$

where $\mathbf{e}_r = \mathbf{r}/r$.

Proof. We have

$$\frac{\partial}{\partial x_i} f(r) = f'(r) \frac{\partial r}{\partial x_i} = f'(r) \frac{x_i}{r}$$
■

Example

Using [proposition 2.5](#) and [corollary 2.11](#), we have

$$\begin{aligned} \nabla \cdot (r^3 \mathbf{r}) &= r^3 \nabla \cdot \mathbf{R} + \mathbf{r} \cdot \nabla r^3 \\ &= 3r^3 + \mathbf{r} \cdot 3r^2 \mathbf{e}_r \\ &= 3r^3 + \mathbf{r} \cdot 3r \mathbf{r} \\ &= 6r^3 \end{aligned}$$

To deal with the modulus of a vector field, we can do

$$\frac{\partial}{\partial x_i} |\mathbf{F}| = \frac{\partial}{\partial x_i} \sqrt{F_j F_j} = \frac{1}{2|\mathbf{F}|} \frac{\partial}{\partial x_i} (F_j F_j) = \frac{1}{|\mathbf{F}|} F_j \frac{\partial F_j}{\partial x_i}$$

§2.8 The Laplacian

Definition. The *Laplacian* of a scalar field is

$$\begin{aligned} \nabla^2 f &= \nabla \cdot (\nabla f) \\ &= \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} \\ &= \frac{\partial^2 f}{\partial x_i \partial x_i} \end{aligned}$$

It may also be denoted Δf .

Example • If $f(\mathbf{x}) = \alpha x^2 + \beta y^2 + \gamma z^2$, then

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) (\alpha x^2 + \beta y^2 + \gamma z^2) = 2(\alpha + \beta + \gamma)$$

- $\nabla^2 |\mathbf{x}|^2 = \nabla^2 (x^2 + y^2 + z^2) = 6$.
- If $f(\mathbf{x}) = (\mathbf{a} \cdot \mathbf{x})^2$,

$$\begin{aligned} \nabla^2 f &= \nabla \cdot (\nabla f) \\ &= \nabla \cdot (2(\mathbf{a} \cdot \mathbf{x}) \nabla (\mathbf{a} \cdot \mathbf{x})) \\ &= \nabla \cdot (2(\mathbf{a} \cdot \mathbf{x}) \mathbf{a}) \\ &= 2(\mathbf{a} \cdot \mathbf{x}) \nabla \cdot \mathbf{a} + 2\mathbf{a} \cdot \nabla (\mathbf{a} \cdot \mathbf{x}) \\ &= 2\mathbf{a} \cdot \mathbf{a} \\ &= 2|\mathbf{a}|^2 \end{aligned}$$

Alternatively, and perhaps more easily,

$$\begin{aligned} \nabla^2 f &= \nabla^2 (a_1^2 x^2 + a_2^2 y^2 + a_3^2 z^2 + 2a_1 a_2 xy + 2a_1 a_3 xz + 2a_2 a_3 yz) \\ &= 2(a_1^2 + a_2^2 + a_3^2) \\ &= 2|\mathbf{a}|^2 \end{aligned}$$

Functions that satisfy $\nabla^2 f = 0$ (Laplace's equation) are called *harmonic*.

We often have spherically symmetric functions, i.e. those for which $f = f(r)$. In that case,

$$\begin{aligned} \nabla^2 f(r) &= \nabla \cdot (\nabla f(r)) \\ &= \nabla \cdot (f'(r) \mathbf{e}_r) \\ &= \frac{\partial}{\partial x_i} \left(\frac{f'(r)}{r} x_i \right) \\ &= \frac{r f''(r) - f'(r)}{r^2} \frac{x_i}{r} x_i + \frac{f'(r)}{r} \delta_{ii} \\ &= f''(r) + \frac{2f'(r)}{r} \end{aligned}$$

This can also be written $\frac{1}{r^2} \frac{d}{dr} (r^2 f'(r))$ or $\frac{1}{r} \frac{d^2}{dr^2} (rf)$.

Remark. As we can see, the only thing that depends on the dimension is δ_{ii} . So, if we were in 2D, we would have $f''(r) + \frac{f'(r)}{r}$ instead.

For example, $\nabla^2 |\mathbf{x}|^2 = \nabla^2 f$ with $f(r) = r^2$, so $\nabla^2 f = 2 + \frac{4r}{r} = 6$.

For a vector field $\mathbf{F}(\mathbf{x})$, we can't define $\nabla^2 \mathbf{F}$ as $\nabla \cdot (\nabla \mathbf{F})$, since $\nabla \mathbf{F}$ is undefined. Instead, we say

$$\nabla^2 \mathbf{F} = (\nabla^2 F_1, \nabla^2 F_2, \nabla^2 F_3)$$

An alternative definition that is coordinate-independent is

$$\nabla^2 \mathbf{F} = \nabla(\nabla \cdot \mathbf{F}) - \nabla \times (\nabla \times \mathbf{F})$$

but the first definition is preferable for calculations. To show they are equivalent, we have

$$\begin{aligned} (\nabla \times (\nabla \times \mathbf{F}))_i &= \varepsilon_{ijk} \frac{\partial}{\partial x_j} \varepsilon_{klm} \frac{\partial F_m}{\partial x_l} \\ &= \frac{\partial^2 F_j}{\partial x_i \partial x_j} - \frac{\partial^2 F_i}{\partial x_j \partial x_j} \\ &= \frac{\partial}{\partial x_i} (\nabla \cdot \mathbf{F}) - \nabla^2 (F_i) \end{aligned}$$

§2.9 Irrotational and conservative vector fields

Consider a vector field $\mathbf{F}$ defined in a domain $\mathcal{D} \subseteq \mathbb{R}^3$.

Definition. If $\mathbf{F}$ satisfies $\nabla \times \mathbf{F} = \mathbf{0}$ in $\mathcal{D}$, it is *irrotational* there.

Definition. If instead there exists a single-valued scalar field f defined in $\mathcal{D}$ such that $\mathbf{F} = \nabla f$, then $\mathbf{F}$ is said to be *conservative*, and f is known as a *scalar potential* for $\mathbf{F}$. Sometimes the sign of f is changed, so $\mathbf{F} = -\nabla f$.

Proposition 2.12

A vector field is conservative if and only if it is irrotational.

Proof. $\implies$ If $\mathbf{F} = \nabla f$, then $\nabla \times \mathbf{F} = \nabla \times (\nabla f)$. But

$$(\nabla \times (\nabla f))_i = \varepsilon_{ijk} \frac{\partial}{\partial x_i} \frac{\partial f}{\partial x_k}$$

and ε_{ijk} is antisymmetric while $\frac{\partial^2 f}{\partial x_j \partial x_k}$ is symmetric, so the terms cancel in pairs.

$\impliedby$ The converse holds only if $\mathcal{D}$ is *simply connected*. The result will be shown in a later section. ■

Definition. A *simply connected domain* $\mathcal{D}$ is a connected domain in which any closed curve can be shrunk continuously to a point.

Example • $\mathbb{R}^3$ is simply connected.

- $\mathbb{R}^3 \setminus \{(x, 0, 0) : x \in \mathbb{R}\}$ is not simply connected - take a curve wrapped around the x -axis.
- $\mathbb{R}^3 \setminus \{(x, 0, 0) : x \geq 0\}$ is simply connected.
- $\mathbb{R}^3 \setminus \{\mathbf{0}\}$ is simply connected.

Importantly, consider $\mathbf{F}(\mathbf{x}) = \frac{1}{x^2+y^2}(-y, x, 0)$. We can see that $\mathbf{F}$ is irrotational. We can also see that $f(\mathbf{x}) = \arctan(y/x)$ satisfies $\mathbf{F} = \nabla f$ - but f is not a single-valued function - or, if we choose the principal value, it's not continuous. Indeed, the domain of $\mathbf{F}$ is not simply connected, since it is undefined on the z -axis.

Lecture n

§2.10 Solenoidal/incompressible vector fields

Definition. A vector field $\mathbf{F}$ is *solenoidal* (or *divergence-free*) in a domain $\mathcal{D}$ if $\nabla \cdot \mathbf{F} = 0$ there.

Remark. The nomenclature arises from magnetic fields, which are always solenoidal (this is actually equivalent to the non-existence of magnetic monopoles). In fluid dynamics, on the other hand, such fields are called incompressible.

If there exists a vector field $\mathbf{A}$ in $\mathcal{D}$ such that $\mathbf{F} = \nabla \times \mathbf{A}$, then $\mathbf{F}$ is certainly solenoidal because

$$\nabla \cdot (\nabla \times \mathbf{A}) = \frac{\partial}{\partial x_i} \varepsilon_{ijk} \frac{\partial A_k}{\partial x_j} = 0$$

Such an $\mathbf{A}$ is called a *vector potential*. The converse also holds in a suitable domain $\mathcal{D}$.

Remark. For the converse to hold, we need $\mathcal{D}$ to be simply connected, and we also need *surfaces* in $\mathcal{D}$ to be able to be shrunk to a point. But this is complicated and non-examinable.

§3 Orthogonal curvilinear coordinate systems

§3.1 General curvilinear coordinate systems

Definition. A *curvilinear coordinate system* in 3D is a parameterisation (u, v, w) of $\mathbb{R}^3$ – in other words, a surjective map $\mathbf{x}(u, v, w)$ from a subset of $\mathbb{R}^3$ to $\mathbb{R}^3$.

A well-known example in $\mathbb{R}^2$ is plane polar coordinates.

Remark. Ideally, the map $\mathbf{x}$ would be invertible, but in practice most coordinate systems fail to be invertible at points – for example, we can't invert at the origin in polar coordinates.

We define the *scale factors*

$$h_u = \left| \frac{\partial \mathbf{x}}{\partial u} \right|, \quad h_v = \left| \frac{\partial \mathbf{x}}{\partial v} \right|, \quad h_w = \left| \frac{\partial \mathbf{x}}{\partial w} \right|$$

and corresponding unit vectors

$$\mathbf{e}_u = \frac{1}{h_u} \frac{\partial \mathbf{x}}{\partial u}, \quad \mathbf{e}_v = \frac{1}{h_v} \frac{\partial \mathbf{x}}{\partial v}, \quad \mathbf{e}_w = \frac{1}{h_w} \frac{\partial \mathbf{x}}{\partial w}$$

Definition. An *orthogonal curvilinear coordinate system* is one in which $\mathbf{e}_u$, $\mathbf{e}_v$, $\mathbf{e}_w$ are mutually orthogonal, and we normally re-order them to form a right-handed set. This allows them to be used as a basis.

Importantly, this basis is *local* – the basis vectors depend on position. For example, a vector field $\mathbf{F}(\mathbf{x})$ can be expressed at $\mathbf{x}$ in terms of its components relative to the basis vectors at $\mathbf{x}$ – so $\mathbf{F} = F_u \mathbf{e}_u + F_v \mathbf{e}_v + F_w \mathbf{e}_w$. But even if $\mathbf{F}$ is a constant vector, *its components won't be*.

By extending [proposition 1.3](#) to the vector function $\mathbf{x}(u, v, w)$, we have the following result:

Proposition 3.1

$$\begin{aligned}
 d\mathbf{x} &= \frac{\partial \mathbf{x}}{\partial u} du + \frac{\partial \mathbf{x}}{\partial v} dv + \frac{\partial \mathbf{x}}{\partial w} dw \\
 &= h_u du \mathbf{e}_u + h_v dv \mathbf{e}_v + h_w dw \mathbf{e}_w
 \end{aligned}$$

whereas in Cartesian coordinates it would simply be $du \mathbf{e}_u + dv \mathbf{e}_v + dw \mathbf{e}_w$. Similarly, the position vector is *not* given by $u \mathbf{e}_u + v \mathbf{e}_v + w \mathbf{e}_w$ – there is a difference between the coordinates (u, v, w) and the position vector $\mathbf{x}(u, v, w)$.

Consider a displacement that keeps u fixed, but allows v and w to vary. Then, from [proposition 3.1](#), $d\mathbf{x}$ is a linear combination of $\mathbf{e}_v$ and $\mathbf{e}_w$, so $d\mathbf{x} \perp \mathbf{e}_u$ for all possible $d\mathbf{x}$. Hence $\mathbf{e}_u$ is perpendicular to any surface of constant u , and similarly for $\mathbf{e}_v$ and $\mathbf{e}_w$. So surfaces of constant u , constant v and constant w all intersect orthogonally at any given point.

§3.2 Cylindrical polar coordinates

Definition. Cylindrical polar coordinates are given by

$$\begin{aligned}
 x &= \rho \cos \phi \\
 y &= \rho \sin \phi \\
 z &= z
 \end{aligned}$$

DIAGRAM! where $\rho \geq 0$ and ϕ can be restricted to $[0, 2\pi)$, but doesn't need to be. We can invert this using

$$\begin{aligned}
 \rho &= \sqrt{x^2 + y^2} \\
 \tan \phi &= y/x \\
 z &= z
 \end{aligned}$$

Then surfaces of constant ρ are cylinders around the z -axis; of constant ϕ are vertical planes through the z -axis; and of constant z are horizontal planes.

Different notations are used – notably r for ρ and θ for ϕ .

We have the following results about cylindrical polar coordinates:

Proposition 3.2

$$\begin{aligned}
 h_\rho &= 1 & h_\phi &= \rho & h_z &= 1 \\
 \mathbf{e}_\rho &= \begin{pmatrix} \cos \phi \\ \sin \phi \\ 0 \end{pmatrix} & \mathbf{e}_\phi &= \begin{pmatrix} -\sin \phi \\ \cos \phi \\ 0 \end{pmatrix} & \mathbf{e}_z &= \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}
 \end{aligned}$$

We can confirm that $\{\mathbf{e}_\rho, \mathbf{e}_\phi, \mathbf{e}_z\}$ is orthonormal and right-handed. From [proposition 3.1](#) we obtain

$$d\mathbf{x} = d\rho \mathbf{e}_\rho + \rho d\phi \mathbf{e}_\phi + dz \mathbf{e}_z$$

and

Proposition 3.3

$$\mathbf{x} = \rho \mathbf{e}_\rho + z \mathbf{e}_z$$

with no $\phi \mathbf{e}_\phi$ – because $\mathbf{e}_\rho$'s already got ϕ in it.

§3.3 Spherical polar coordinates

Definition. *Spherical polar coordinates* are given by

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

where $r \geq 0$, $\theta \in [0, \pi]$, and ϕ can be restricted to $[0, 2\pi)$ – DIAGRAM!! We can invert this using

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$\tan \theta = \frac{\sqrt{x^2 + y^2}}{z}$$

$$\tan \phi = y/x$$

Then surfaces of constant r are spheres centred at the origin; of constant θ are cones with apex at the origin; and of constant ϕ are vertical planes through the z -axis.

Remark. Some people order the coordinates as (r, ϕ, θ) instead. Some swap the definitions of ϕ and θ . Some do both. Some replace θ with $\frac{\pi}{2} - \theta$. This can be extremely confusing.

Proposition 3.4

For spherical polar coordinates,

$$h_r = 1$$

$$h_\theta = r$$

$$h_\phi = r \sin \theta$$

$$\mathbf{e}_r = \begin{pmatrix} \sin \theta \cos \phi \\ \sin \theta \sin \phi \\ \cos \theta \end{pmatrix}$$

$$\mathbf{e}_\theta = \begin{pmatrix} \cos \theta \cos \phi \\ \cos \theta \sin \phi \\ -\sin \theta \end{pmatrix}$$

$$\mathbf{e}_\phi = \begin{pmatrix} -\sin \phi \\ \cos \phi \\ 1 \end{pmatrix}$$

We also have

Proposition 3.5

$$\mathbf{x} = r \mathbf{e}_r$$

Finally, from [proposition 3.1](#) we obtain

$$d\mathbf{x} = dr \mathbf{e}_r + r d\theta \mathbf{e}_\theta + r \sin \theta d\phi \mathbf{e}_\phi$$

§3.4 Vector differential operators in orthogonal curvilinear coordinate systems

§3.4.1 General orthogonal curvilinear coordinate systems

If we have a function expressed in terms of u, v, w , i.e. in arbitrary orthogonal curvilinear coordinates, we would like to be able to express ∇f in terms of u, v, w as well. Observe firstly that

$$\nabla f \cdot \mathbf{e}_u = \frac{\partial f}{\partial x_i} \frac{1}{h_u} \frac{\partial x_i}{\partial u} = \frac{1}{h_u} \frac{\partial f}{\partial u}$$

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Since $\{\mathbf{e}_u, \mathbf{e}_v, \mathbf{e}_w\}$ form an orthonormal set, doing these for each of the basis vectors will give us the components of ∇f along each basis vector. We have

$$\nabla f = \frac{1}{h_u} \frac{\partial f}{\partial u} \mathbf{e}_u + \frac{1}{h_v} \frac{\partial f}{\partial v} \mathbf{e}_v + \frac{1}{h_w} \frac{\partial f}{\partial w} \mathbf{e}_w$$

and furthermore, we can take out the f to get

Proposition 3.6

$$\nabla = \frac{1}{h_u} \mathbf{e}_u \frac{\partial}{\partial u} + \frac{1}{h_v} \mathbf{e}_v \frac{\partial}{\partial v} + \frac{1}{h_w} \mathbf{e}_w \frac{\partial}{\partial w}$$

Calculating the divergence, curl and Laplacian of a vector field in general orthogonal curvilinear coordinates is extremely messy. There are a range of ways to do it – we will see the best method in section 6.5.

ADD THEM IN

§3.4.2 Cylindrical polar coordinates

For a scalar field $f(\rho, \phi, z)$ and vector field $F_\rho \mathbf{e}_\rho + F_\phi \mathbf{e}_\phi + F_z \mathbf{e}_z$, we obtain

$$\begin{aligned} \nabla f &= \frac{\partial f}{\partial \rho} \mathbf{e}_\rho + \frac{1}{\rho} \frac{\partial f}{\partial \phi} \mathbf{e}_\phi + \frac{\partial f}{\partial z} \mathbf{e}_z \\ \nabla \cdot \mathbf{F} &= \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho F_\rho) + \frac{1}{\rho} \frac{\partial F_\phi}{\partial \phi} + \frac{\partial F_z}{\partial z} \\ \nabla \times \mathbf{F} &= \left(\frac{1}{\rho} \frac{\partial F_z}{\partial \phi} - \frac{\partial F_\phi}{\partial z} \right) \mathbf{e}_\rho + \left(\frac{\partial F_\rho}{\partial z} - \frac{\partial F_z}{\partial \rho} \right) \mathbf{e}_\phi + \frac{1}{\rho} \left(\frac{\partial}{\partial \rho} (\rho F_\phi) - \frac{\partial F_\rho}{\partial \phi} \right) \mathbf{e}_z \\ \nabla^2 f &= \frac{\partial^2 f}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial f}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 f}{\partial \phi^2} + \frac{\partial^2 f}{\partial z^2} \end{aligned}$$

It is required to know ∇f and the radial terms of $\nabla^2 f$ by heart. It is not necessary to learn div and curl.

Usefully, if we want to get the operators for plane polar coordinates, we can just throw out all terms involving z .

§3.4.3 Spherical polar coordinates

For $f(r, \theta, \phi)$ and $F_r \mathbf{e}_r + F_\theta \mathbf{e}_\theta + F_\phi \mathbf{e}_\phi$, we have

$$\begin{aligned}\nabla f &= \frac{\partial f}{\partial r} \mathbf{e}_r + \frac{1}{r} \frac{\partial f}{\partial \theta} \mathbf{e}_\theta + \frac{1}{r \sin \theta} \frac{\partial f}{\partial \phi} \mathbf{e}_\phi \\ \nabla \cdot \mathbf{F} &= \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 F_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (F_\theta \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial F_\phi}{\partial \phi} \\ \nabla \times \mathbf{F} &= \frac{1}{r \sin \theta} \left(\frac{\partial}{\partial \theta} (F_\phi \sin \theta) - \frac{\partial F_\theta}{\partial \phi} \right) \mathbf{e}_r + \\ &\quad \frac{1}{r} \left(\frac{1}{\sin \theta} \frac{\partial F_r}{\partial \phi} - \frac{\partial}{\partial r} (r F_\phi) \right) \mathbf{e}_\theta + \\ &\quad \frac{1}{r} \left(\frac{\partial}{\partial r} (r F_\theta) - \frac{\partial F_r}{\partial \theta} \right) \mathbf{e}_\phi \\ \nabla^2 f &= \frac{\partial^2 f}{\partial r^2} + \frac{2}{r} \frac{\partial f}{\partial r} + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\frac{\partial f}{\partial \theta} \sin \theta \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 f}{\partial \phi^2}\end{aligned}$$

Again, it is required to know only ∇f and the radial terms of $\nabla^2 f$ by heart.

Example • Let $f(\mathbf{x}) = |\mathbf{x}|^2$ and $\mathbf{F}(\mathbf{x}) = \mathbf{x}$. We have already obtained $\nabla f = 2\mathbf{x}$ and $\nabla \cdot \mathbf{F} = 3$ using Cartesian coordinates.

Instead working in cylindricals, we have $\mathbf{x} = \rho \mathbf{e}_\rho + z \mathbf{e}_z$, so $f(\rho, \phi, z) = \rho^2 + z^2$, while $F_\rho = \rho$, $F_\phi = 0$, $F_z = z$. Now we can apply the formulae:

$$\begin{aligned}\nabla f &= \frac{\partial f}{\partial \rho} \mathbf{e}_\rho + \frac{1}{\rho} \frac{\partial f}{\partial \phi} \mathbf{e}_\phi + \frac{\partial f}{\partial z} \mathbf{e}_z = 2\rho \mathbf{e}_\rho + 2z \mathbf{e}_z = 2\mathbf{x} \\ \nabla \cdot \mathbf{F} &= \frac{1}{\rho} \frac{\partial}{\partial \rho}(\rho F_\rho) + \frac{1}{\rho} \frac{\partial F_\phi}{\partial \phi} + \frac{\partial F_z}{\partial z} = 2 + 1 = 3\end{aligned}$$

- In sphericals, we have $\mathbf{x} = r \mathbf{e}_r$, so $f(r, \theta, \phi) = r^2$, while $F_r = r$, $F_\theta = 0$, $F_\phi = 0$. Hence

$$\begin{aligned}\nabla f &= \frac{\partial f}{\partial r} \mathbf{e}_r = 2r \mathbf{e}_r = 2\mathbf{x} \\ \nabla \cdot \mathbf{F} &= \frac{1}{r^2} \frac{\partial}{\partial r}(r^2 F_r) = \frac{1}{r^2} \frac{\partial}{\partial r}(r^3) = 3\end{aligned}$$

- Consider the gravitational force field $\mathbf{F} = -(GMm/r^2) \mathbf{e}_r$ of a fixed mass M at the origin acting on another mass m . We have

$$\nabla \cdot \mathbf{F} = -\frac{1}{r^2} \frac{\partial}{\partial r}(r^2 \cdot \frac{GMm}{r^2}) = 0$$

- The following simple example shows how we could obtain the expressions given above. Consider a simplified field of the form $\mathbf{F} = \alpha(\theta) \mathbf{e}_\theta$. We have

$$\begin{aligned}\nabla \times \mathbf{F} &= \left(\frac{\partial}{\partial r} \mathbf{e}_r + \frac{1}{r} \frac{\partial}{\partial \theta} \mathbf{e}_\theta + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} \mathbf{e}_\phi \right) \times (\alpha(\theta) \mathbf{e}_\theta) \\ &= \mathbf{e}_r \times \alpha(\theta) \frac{\partial \mathbf{e}_\theta}{\partial r} + \frac{1}{r} \mathbf{e}_\theta \times \frac{\partial}{\partial \theta} (\alpha(\theta) \mathbf{e}_\theta) + \frac{1}{r \sin \theta} \mathbf{e}_\phi \times \alpha(\theta) \frac{\partial \mathbf{e}_\theta}{\partial \phi} \\ &= \dots\end{aligned}$$

§4 Differential geometry of curves, surfaces and volumes

§4.1 Curves and arc length

Definition. A *curve* C is a map $\mathbf{x} : [a, b] \subset \mathbb{R} \rightarrow \mathbb{R}^3$, in other words $\mathbf{x}(t) = (x(t), y(t), z(t))$ for $a \leq t \leq b$. We say that the curve is *parameterised* by t . A curve is *closed* if $\mathbf{x}(a) = \mathbf{x}(b)$.

In this course we will require that curves be continuous and piecewise smooth (infinitely differentiable everywhere except a finite number of points). DIAGRAM: mostly smooth but a few spiky points.

Normally, we will use the word curve to refer to the *image* of the map C , not the map itself.

Definition. The *arc length* of a curve is given by

$$s = \int ds = \int |\dot{\mathbf{x}}(t)| dt$$

This is motivated by

$$\begin{aligned}\delta s = |\delta \mathbf{x}| &= \sqrt{\delta x^2 + \delta y^2 + \delta z^2} \\ &= \sqrt{\left(\frac{\delta x}{\delta t}\right)^2 + \left(\frac{\delta y}{\delta t}\right)^2 + \left(\frac{\delta z}{\delta t}\right)^2} \\ &= \left|\frac{\delta \mathbf{x}}{\delta t}\right| \delta t\end{aligned}$$

which, taking the limit, gives

Proposition 4.1

$$\frac{ds}{dt} = |\dot{\mathbf{x}}|$$

Thus we obtain s as a function of t which we can, at least theoretically, invert to reparameterise a curve in terms of its arc length, which could be quite useful – the so-called *natural parameterisation*.

- Given two curves C_1 and C_2 with matching end-points, we define the curve $C_1 + C_2$ to be the two curves joined end-to-end.
- The curve $-C$ is the curve C traversed in the opposite direction – $\mathbf{x}_{-C}(t) = \mathbf{x}_C(a + b - t)$.

Example • Consider $\mathbf{x}(t) = (at^2, 2at, 0)$, $t \in [-1, 1]$. This is a parameterisation (the standard one) of the parabola $y^2 = 4ax$ between $y = \pm 2a$.

- $\mathbf{x}(t) = (a \cos t, b \sin t, 0)$, $t \in [0, \pi]$ is a parameterisation of the upper half of the ellipse $x^2/a^2 + y^2/b^2 = 1$.
- $\mathbf{x}(t) = (\cos t, \sin t, t)$ is a helix. In cylindrical polars, $\rho = 1$, $\phi = t$, $z = t$, in other words the curve winds its way around the unit cylinder.

When drawing our curves, we should add an arrow to indicate the direction in which t increases.

Example (Bonus non-examinable!)

Let $R(t) = 2 + \cos \alpha t$ and $\mathbf{x}(t) = (R(t) \cos \beta t, R(t) \sin \beta t, \sin \alpha t)$. This curve winds its way around the torus given in cylindrical polars by $(\rho - 2)^2 + z^2 = 1$. If $\alpha/\beta \in \mathbb{Q}$, the curve closes on itself – in fact, if $\alpha/\beta = p/q$ where p and q are coprime integers, then we have the *torus knot* $K_{q,p}$. But if $\alpha/\beta \notin \mathbb{Q}$, the curve covers the entire torus.

§4.2 The Frenet-Serret equations

We will assume in this section that $\mathbf{x}$ is parameterised by arc length s . If it is not, then the following results *do not apply*. For clarity, we will use primes to denote differentiation with respect to this canonical parameterisation. We define

Proposition 4.2

$$\begin{array}{ll}
\mathbf{t} = \mathbf{x}' & \text{the } \textit{tangent} \text{ vector} \\
\mathbf{n} = \frac{1}{|\mathbf{x}''|} \mathbf{x}'' & \text{the } \textit{principal normal} \text{ vector} \\
\mathbf{b} = \mathbf{t} \times \mathbf{n} & \text{the } \textit{binormal} \text{ vector}
\end{array}$$

Proposition 4.3 (Frenet-Serret equations)

$$\begin{aligned}
\mathbf{t}' &= \kappa \mathbf{n} \\
\mathbf{n}' &= \tau \mathbf{b} - \kappa \mathbf{t} \\
\mathbf{b}' &= -\tau \mathbf{n}
\end{aligned}$$

where $\kappa = |\mathbf{x}''|$ and $\tau = -\mathbf{b}' \cdot \mathbf{n}$ are scalars, not necessarily constant, called the *curvature* and *torsion* of the curve respectively.

Proof. First, $ds = |d\mathbf{x}|$, so $|\frac{d\mathbf{x}}{ds}| = 1$ – hence $|\mathbf{t}| = 1$. Now $\kappa = |\mathbf{x}''| = |\mathbf{t}'|$, so $\mathbf{n} = \mathbf{t}'/\kappa$ by definition. Hence $|\mathbf{n}| = 1$ and $\mathbf{t}' = \kappa \mathbf{n}$ (later on we'll worry about $\kappa = 0$).

Differentiating $\mathbf{t} \cdot \mathbf{t} = 1$ gives $2\mathbf{t} \cdot \mathbf{t}' = 0$, so $\mathbf{t} \cdot \mathbf{n} = 0$, that is $\mathbf{t} \perp \mathbf{n}$. Then, by the definition $\mathbf{b} = \mathbf{t} \times \mathbf{n}$, we can be sure $\{\mathbf{t}, \mathbf{n}, \mathbf{b}\}$ is orthonormal and right-handed. This set (which depends on the position along the curve) is known as the *moving trihedral*.

Now $\mathbf{b} \cdot \mathbf{b} = 1$, so $\mathbf{b} \cdot \mathbf{b}' = 0$, and $\mathbf{t} \cdot \mathbf{b} = 0$, so

$$\begin{aligned}
0 &= \mathbf{t}' \cdot \mathbf{b} + \mathbf{t} \cdot \mathbf{b}' \\
&= \kappa \mathbf{n} \cdot \mathbf{b} + \mathbf{t} \cdot \mathbf{b}' \\
&= \mathbf{t} \cdot \mathbf{b}'
\end{aligned}$$

Therefore $\mathbf{b}' \perp \mathbf{t}$ and $\mathbf{b}' \perp \mathbf{b}$, so it must be that $\mathbf{b}' \parallel \mathbf{n}$ and, from the definition of τ , $\mathbf{b}' = -\tau \mathbf{n}$.

Finally, $\mathbf{n} = \mathbf{b} \times \mathbf{t}$, so

$$\mathbf{n}' = \mathbf{b}' \times \mathbf{t} + \mathbf{b} \times \mathbf{t}' = \tau \mathbf{n} \times \mathbf{t} + \mathbf{b} \times \kappa \mathbf{n} = \tau \mathbf{b} - \kappa \mathbf{t}$$

as required.

Let's now worry about $\kappa = 0$. In that case, $\mathbf{n}$, $\mathbf{b}$ and τ are also all undefined. If $\kappa = 0$ everywhere, the curve is just a straight line. ■

Definition. The *radius of curvature* is defined as $R = \kappa^{-1}$.

At a point $\mathbf{x}$, the *osculating plane* is the plane that passes through $\mathbf{x}$ and is spanned by $\mathbf{t}$ and $\mathbf{n}$. Then a circle of radius R centred at $\mathbf{x} + R\mathbf{n}$ is the best possible approximation of the curve at $\mathbf{x}$ by a circle – it has the same values of $\mathbf{x}$, $\mathbf{x}'$ and $\mathbf{x}''$ as the curve itself (as well as κ , of course).

DIAGRAM: diagram.

§4.3 Using the Frenet-Serret equations with other parameterisations

One option is always to simply reparameterise our curve in terms of s . But this is not really necessary. Instead, we can simply apply the chain rule:

$$\begin{aligned}\mathbf{t} &= \mathbf{x}' = \frac{d\mathbf{x}}{dt} \frac{dt}{ds} = \frac{\dot{\mathbf{x}}}{|\dot{\mathbf{x}}|} \\ \mathbf{x}'' &= \frac{d\mathbf{t}}{ds} = \frac{d\mathbf{t}}{dt} \frac{dt}{ds} = \frac{\dot{\mathbf{t}}}{|\dot{\mathbf{x}}|}\end{aligned}$$

so that $\kappa = \frac{|\dot{\mathbf{t}}|}{|\dot{\mathbf{x}}|}$, $\mathbf{n} =$

Example

Consider $\mathbf{x}(t) = (\frac{1}{\sqrt{2}}t^2, t, t)$. Then $\dot{\mathbf{x}} = (\sqrt{2}t, 1, 1)$, so $\frac{ds}{dt} = |\dot{\mathbf{x}}| = \sqrt{2}\sqrt{1+t^2}$. Hence

$$\begin{aligned}\mathbf{t} &= \mathbf{x}' = \frac{\dot{\mathbf{x}}}{|\dot{\mathbf{x}}|} = \frac{1}{\sqrt{2}\sqrt{1+t^2}}(\sqrt{2}t, 1, 1) \\ \dot{\mathbf{t}} &= \frac{1}{\sqrt{2}(1+t^2)^{3/2}}(\sqrt{2}, -t, -t) \\ \kappa &= |\mathbf{t}'| = \frac{|\dot{\mathbf{t}}|}{|\dot{\mathbf{x}}|} = \frac{1}{\sqrt{2}(1+t^2)^{3/2}}\end{aligned}$$

DIAGRAM: diagram

§4.4 Surfaces, surface elements and the normal

Definition. A *surface* is a map $\mathbf{x} : U \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}^3$. We typically use u and v to refer to the arguments of $\mathbf{x}$.

As with curves, we will normally refer to the image of the map, not the map itself.

Example

We might wish to make a hemispherical surface, $H_R = \{\mathbf{x} : |\mathbf{x}| = R, z < 0\}$. We can use spherical polar coordinates to parameterise this as

$$\mathbf{x}(\theta, \phi) = \begin{pmatrix} R \sin \theta \cos \phi \\ R \sin \theta \sin \phi \\ R \cos \theta \end{pmatrix}$$

with $\theta \in (\pi/2, \pi]$, $\phi \in [0, 2\pi)$.

Lemma

Given a parallelogram with two sides $\mathbf{a}$ and $\mathbf{b}$, its area is $|\mathbf{a} \times \mathbf{b}|$.

Consider now a region lying between curves of constant u and curves of constant v . What is the area of such a region?

Suppose we have two such curves meeting at $\mathbf{x}(u, v)$, and nearby curves at constant $u + \delta u$ and $v + \delta v$. The part of S bounded by these curves forms, at lowest order, a very small parallelogram, with sides given by

$$\begin{aligned}\mathbf{x}(u + \delta u, v) - \mathbf{x}(u, v) &= \frac{\partial \mathbf{x}}{\partial u} \delta u \\ \mathbf{x}(u, v + \delta v) - \mathbf{x}(u, v) &= \frac{\partial \mathbf{x}}{\partial v} \delta v\end{aligned}$$

Hence, by the lemma, the area of the region is given by

Proposition 4.4

$$dS = \left| \frac{\partial \mathbf{x}}{\partial u} \times \frac{\partial \mathbf{x}}{\partial v} \right| du dv$$

This infinitesimal parallelogram is referred to as a *surface element*.

Remark. Importantly, the normal to the surface at $\mathbf{x}$ is parallel to $\frac{\partial \mathbf{x}}{\partial u} \times \frac{\partial \mathbf{x}}{\partial v}$.

§4.5 Volumes and volume elements

Definition. A *volume* is a map $\mathbf{x} : U \subseteq \mathbb{R}^3 \rightarrow \mathbb{R}^3$. We typically use u , v and w to refer to the arguments of $\mathbf{x}$.

Example

To parameterise a sphere, we can do a similar thing to before, but let $r \in [0, R]$.

Similarly, we can divide our volumes into regions lying between *surfaces* of constant u , v and w . To lowest order, these regions are *parallelepipeds* with sides $\frac{\partial \mathbf{x}}{\partial u} \delta u$, $\frac{\partial \mathbf{x}}{\partial v} \delta v$ and $\frac{\partial \mathbf{x}}{\partial w} \delta w$. Hence the volume element is given by

Proposition 4.5

$$dV = \left| \frac{\partial \mathbf{x}}{\partial u} \cdot \left(\frac{\partial \mathbf{x}}{\partial v} \times \frac{\partial \mathbf{x}}{\partial w} \right) \right| du dv dw$$

§5 Line, area, surface and volume integrals

§5.1 Line integrals

Definition. Given a curve C parameterised by $t \in [a, b]$, the *line integral* of a vector field $\mathbf{F}$ along C is

$$\int_C \mathbf{F} \cdot d\mathbf{x} = \int_a^b \mathbf{F}(\mathbf{x}(t)) \cdot \frac{d\mathbf{x}}{dt} dt$$

An alternative but equivalent definition is to return to a Riemann sum: we define

$$\int_C \mathbf{F} \cdot d\mathbf{x} = \lim_{\Delta \rightarrow 0} \sum_{n=0}^{N-1} \mathbf{F}(\mathbf{x}_n) \cdot \delta \mathbf{x}_n$$

Remark. We can also take a line integral on a scalar field, or take a line integral while crossing (!) $\mathbf{F}$ with $d\mathbf{x}$ instead of dotting.

If C is a closed curve, then the line integral around the whole curve has a special notation

$$\oint_C \mathbf{F} \cdot d\mathbf{x}$$

and this is sometimes known as the *circulation* of $\mathbf{F}$ around C . It also doesn't matter where we start along such a curve, so long as we go all the way around.

We have some elementary properties of line integrals:

- $\int_{-C} \mathbf{F} \cdot d\mathbf{x} = - \int_C \mathbf{F} \cdot d\mathbf{x};$
- $\int_{C_1+C_2} \mathbf{F} \cdot d\mathbf{x} = \int_{C_1} \mathbf{F} \cdot d\mathbf{x} + \int_{C_2} \mathbf{F} \cdot d\mathbf{x}.$

Example • Consider $\mathbf{F} = (y, -x, 0)$, and C the straight line joining $(-1, 0, 0)$ to $(1, 0, 0)$. Parameterise the curve as $\mathbf{x} = (t, 0, 0)$ for $t \in [-1, 1]$. Then $\mathbf{F}(\mathbf{x}(t)) = (0, -t, 0)$, and $\dot{\mathbf{x}} = (1, 0, 0)$, so $\int_{C_1} \mathbf{F} \cdot d\mathbf{x} = 0$.

- Consider $\mathbf{F} = (3x^2, 2y, 0)$ on the same line. We obtain $\mathbf{F} \cdot \dot{\mathbf{x}} = (3t^2, 0, 0) \cdot (1, 0, 0)$ and so $\int_{C_1} \mathbf{F} \cdot d\mathbf{x} = \int_{-1}^1 3t^2 dt = 2$.
- Consider $\mathbf{F} = (y, -x, 0)$ along the curve C_2 joining $(-1, 0, 0)$ to $(1, 0, 0)$ parameterised as $\mathbf{x} = (t-1, t(t-2), t(t^2-4))$ for $t \in [0, 2]$. We have

$$\mathbf{F} = (t(t-2), 1-t, 0) \quad \mathbf{x} = (1, 2t-2, 3t^2-4)$$

so

$$\begin{aligned} \int_{C_2} \mathbf{F} \cdot d\mathbf{x} &= \int_0^2 t(t-2) - 2(t-1)^2 dt \\ &= -\frac{8}{3} \end{aligned}$$

- Consider $\mathbf{F} = (3x^2, 2y, 0)$ along C_2 . We have

$$\begin{aligned} \int_{C_2} \mathbf{F} \cdot d\mathbf{x} &= \int_0^2 (3(t-1)^2, 2t(t-2), 0) \cdot (1, 2t-2, 3t^2-4) dt \\ &= \int_0^2 4t^3 - 9t^2 + 2t - 3 dt \\ &= 2 \end{aligned}$$

- What is the work done by a force field $\mathbf{F} = (y^2 e^x, e^y + 2ye^x, 0)$ moving a particle along C_3 , which is the ellipse $x^2/a^2 + y^2/b^2 = 1$ in the (x, y) -plane starting at $(a, 0, 0)$ and going anticlockwise to $(0, b, 0)$?

First, we need to parameterise the curve by $\mathbf{x}(\theta) = (a \cos \theta, b \sin \theta, 0)$ for $\theta \in [0, \pi/2]$. Then

$$\mathbf{F}(\mathbf{x}(\theta)) = (b^2 \sin^2 \theta e^{a \cos \theta}, e^{b \sin \theta} + 2b \sin \theta e^{a \cos \theta}, 0)$$

so that

$$\mathbf{F} \cdot \frac{d\mathbf{x}}{d\theta} = -ab^2 \sin^3 \theta e^{a \cos \theta} + b \cos \theta e^{b \sin \theta} + 2b^2 \sin \theta \cos \theta e^{a \cos \theta}$$

so

$$\begin{aligned} \int_{C_3} \mathbf{F} \cdot d\mathbf{x} &= [e^{b \sin \theta}]_0^{\pi/2} + \int_0^{\pi/2} b^2 \sin \theta e^{a \cos \theta} (-a + 2 \cos \theta + a \cos^2 \theta) d\theta \\ &= e^b - 1 + \int_1^0 b^2 e^{au} (a^2 - 2u - au^2) du \quad (u = \cos \theta) \\ &= e^b - 1 + [b^2 e^{au} (1 - u^2)]_1^0 \\ &= e^b - 1 + b^2 \end{aligned}$$

Around the whole ellipse, starting at $(a, 0, 0)$ and going anticlockwise all the way round, we would find $\oint \mathbf{F} \cdot d\mathbf{x} = 0$.

Example

Let's do one in cylindrical polar coordinates! Consider $\mathbf{F} = z\mathbf{e}_\phi$ on the path $\rho = z^2$, $\phi = \pi z$ from $z = 0$ to $z = 1$. Here, we have

$$\begin{aligned} d\mathbf{x} &= d\rho \mathbf{e}_\rho + \rho d\phi \mathbf{e}_\phi + dz \mathbf{e}_z \\ &= (2z\mathbf{e}_\rho + \pi\rho\mathbf{e}_\phi + \mathbf{e}_z) dz \end{aligned}$$

and, on the curve, this gives $\mathbf{F} \cdot d\mathbf{x} = \pi\rho z dz = \pi z^3 dz$. Hence

$$\int \mathbf{F} \cdot d\mathbf{x} = \int_0^1 \pi z^3 dz = \pi/4$$

§5.2 Path independence

Definition (Exact differential forms – reminder). We say that a differential form $\mathbf{F} \cdot d\mathbf{x}$ is *exact* if there exists a single-valued scalar field $f(\mathbf{x})$ such that $\mathbf{F} \cdot d\mathbf{x} = df$ for all $d\mathbf{x}$.

From [proposition 2.2](#), we know that $df = \nabla f \cdot d\mathbf{x}$ – so $\mathbf{F} \cdot d\mathbf{x}$ is exact if and only if there exists a single-valued function f such that $\mathbf{F} = \nabla f$, i.e. if $\mathbf{F}$ is a conservative vector field.

If $\mathbf{F}$ is a conservative vector field, and C is a path from $\mathbf{a}$ to $\mathbf{b}$, then

$$\int_C \mathbf{F} \cdot d\mathbf{x} = \int_C df = f(\mathbf{b}) - f(\mathbf{a})$$

and this is notably independent of the particular path. So we can, if we like, write

$$\int_{\mathbf{a}}^{\mathbf{b}} \mathbf{F} \cdot d\mathbf{x}$$

In particular, we have the important result that, if C is a closed curve,

$$\oint_C \mathbf{F} \cdot d\mathbf{x} = 0$$

for any conservative $\mathbf{F}$, since $\mathbf{b} = \mathbf{a}$ and f is single-valued.

Now we have the awesome result for vector fields $\mathbf{F}$ in a simply-connected domain:

- ① $\mathbf{F}$ is a conservative vector field;
- ② there exists a single-valued function $f(\mathbf{x})$ such that $\mathbf{F} = \nabla f$;
- ③ $\mathbf{F} \cdot d\mathbf{x}$ is an exact differential;
- ④ $\mathbf{F}$ is irrotational;
- ⑤ $\nabla \times \mathbf{F} = \mathbf{0}$;
- ⑥ $\frac{\partial F_i}{\partial x_j} = \frac{\partial F_j}{\partial x_i}$ for all i, j ;
- ⑦ for any $\mathbf{a}, \mathbf{b}$ in the domain, $\int_{\mathbf{a}}^{\mathbf{b}} \mathbf{F} \cdot d\mathbf{x}$ is path-independent;
- ⑧ $\oint_C \mathbf{F} \cdot d\mathbf{x} = 0$ for all closed curves C in the domain.

To prove ⑧ $\implies$ ⑦ and ⑦ $\implies$ ② is non-trivial and we'll have to wait until section 6.4.

Example

Now we can revisit some of our earlier examples with a new zest. Observe

- for the field $\mathbf{F} = (y, -x, 0)$, we have $\nabla \times \mathbf{F} = (0, 0, -2) \neq \mathbf{0}$, so $\mathbf{F}$ is not conservative - indeed, we obtained different values for different paths.
- for the field $\mathbf{F} = (3x^2, 2y, 0)$, we have $\nabla \times \mathbf{F} = 0$, or $\mathbf{F} = \nabla(x^3 + y^2)$, and indeed we obtained the same answers for both paths.
- for the field $\mathbf{F} = (y^2 e^x, e^y + 2y e^x, 0)$, we have $\mathbf{F} = \nabla(e^y + y^2 e^x)$, so we can avoid all the integration and directly obtain

$$\int_{C_3} \mathbf{F} \cdot d\mathbf{x} = [e^y + y^2 e^x]_{(a,0,0)}^{(0,b,0)} = e^b + b^2 - 1$$

- for the field $\mathbf{F} = 0\mathbf{e}_\rho + z\mathbf{e}_\phi + 0\mathbf{e}_z$, we have $\nabla \times (z\mathbf{e}_\phi) = -\mathbf{e}_\rho \neq 0$. But in fact, even if $\nabla \times \mathbf{F}$ had vanished, we still could not have concluded that $\mathbf{F}$ is conservative, because $\mathbf{e}_\phi$ is undefined on the z -axis and so the domain is not simply connected.

§5.3 Area integrals

To integrate over a region $A \subset \mathbb{R}^2$, we cover A with a large number of small disjoint subsets δA_i . Then

$$\int_A f(x, y) dA = \lim_{\Delta \rightarrow 0} \sum_{i=1}^N f(x_i, y_i) \delta A_i$$

where in particular Δ is the maximum linear dimension of any of the δA_i .

Informally, we can think of $\int_A f(x, y) dA$ as the volume under the graph of $z = f(x, y)$, in analogy with the 1D case.

Note that A refers to the region, as well as its area: we have

$$A = \int_A 1 dA$$

The most obvious way to divide A up is with rectangles parallel to the x and y axes, each with area $\delta x \delta y$. Summing over a horizontal strip with a fixed value of y , with constant height δy and letting $\delta x \rightarrow 0$, we obtain

$$\left(\int_{X(y)} f(x, y) dx \right) \delta y$$

where, for each y , $X(y) = \{x : (x, y) \in A\}$. We then sum these over the range of y , taking $\delta y \rightarrow 0$, to obtain

$$\int_{y_{\min}}^{y_{\max}} \left(\int_{X(y)} f(x, y) dx \right) dy$$

But we could have equally well done this the other way to obtain

$$\int_{x_{\min}}^{x_{\max}} \left(\int_{Y(x)} f(x, y) dy \right) dx$$

Proposition (Fubini's theorem)

For any reasonable function f over a reasonable A , both of these are equal to $\int_A f(x, y) dA$.

Remark. Only very basic conditions are needed. For example, if f is piecewise continuous and the boundary of A is a finite piecewise smooth curve, then Fubini's theorem holds.

Other notations for $\int_A f(x, y) dA$ include $\int_A f(x, y) dx dy$, $\int_A f(\mathbf{x}) d^2\mathbf{x}$, $\iint_A f(x, y) dA$, $\int_A f(x, y) dS$ and even $\int dx \int dy f(x, y)$.

Example

Let us integrate $f(x, y) = xy^2$ over A , a triangle with vertices at $(0, 0)$, $(1, 1)$ and $(1, -1)$ (DIAGRAM). We have

$$\begin{aligned} \int_A f(x, y) dA &= \int_0^1 \int_{-x}^x xy^2 dy dx \\ &= \int_0^1 x \left[\frac{1}{3} y^3 \right]_{-x}^x dx \\ &= \int_0^1 \frac{2}{3} x^4 dx = \frac{2}{15} \end{aligned}$$

Alternatively, we could have done

$$\begin{aligned} \int_A xy^2 dA &= \int_{-1}^1 \int_{|y|}^1 1xy^2 dx dy \\ &= \int_{-1}^1 y^2 \left[\frac{1}{2} x^2 \right]_{|y|}^1 dy \\ &= \int_{-1}^1 \frac{1}{2} y^2 (1 - y^2) dy \\ &= \frac{2}{15} \end{aligned}$$

Importantly, the inside integral should have limits specified in terms of x or y .

In the special case that A is a rectangle $[a, b] \times [c, d]$ and f is separable, i.e. $f(x, y) = g(x)h(y)$, then

$$\int_A f(x, y) dA = \left(\int_a^b g(x) dx \right) \left(\int_c^d h(y) dy \right)$$

§5.4 Change of variables for area integrals

Proposition

Let $(u, v) \mapsto (x, y)$ be a bijection from a region A' in the (u, v) -plane to a region A in the (x, y) -plane. Then

$$\int_A f(x, y) \, dx \, dy = \int_{A'} F(u, v) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| \, du \, dv$$

where $F(u, v) = f(x(u, v), y(u, v))$ and

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \det \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix}$$

is called the *Jacobian* of the transformation. Importantly, it is the *modulus* of the Jacobian that appears in the integral.

Remark. We often call capital F little f instead, although this is an obvious abuse of notation.

Proof. Let us divide A up using curves of constant u and v . The area of each region is given by

$$\begin{aligned} \left| \frac{\partial \mathbf{x}}{\partial u} \times \frac{\partial \mathbf{x}}{\partial v} \right| \, du \, dv &= \left| \begin{pmatrix} \frac{\partial x}{\partial u} \\ \frac{\partial y}{\partial u} \\ 0 \end{pmatrix} \times \begin{pmatrix} \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial v} \\ 0 \end{pmatrix} \right| \, du \, dv \\ &= \left| \frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial x}{\partial v} \frac{\partial y}{\partial u} \right| \, du \, dv \\ &= |J| \, du \, dv \end{aligned}$$

Then we simply need to add up all the infinitesimal areas $f(\mathbf{x}(u, v))|J| \, du \, dv$. ■

Remark. A full proof of this result requires first a proof of the cross product expression for the infinitesimal area.

Example

Consider $\int_A y^2 dA$ where A is the semi-circular domain $\{(x, y) : x^2 + y^2 < R^2, y > 0\}$. We choose plane polar coordinates (r, θ) with $x = r \cos \theta$, $y = r \sin \theta$. Then, we have a bijection between $A' = \{(r, \theta) : r < R, 0 < \theta < \pi\}$, so

$$\frac{\partial(x, y)}{\partial(r, \theta)} = \det \begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{pmatrix} = \det \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix} = r$$

and so

$$\begin{aligned} \int_A y^2 dA &= \int_{A'} r^2 \sin^2 \theta |r| dr d\theta \\ &= \int_0^R r^3 dr \int_0^\pi \sin^2 \theta d\theta \\ &= \frac{1}{8} \pi R^4 \end{aligned}$$

DIAGRAM of the area element.

§5.5 Properties of the Jacobian